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Problem Statement:

Urban traffic congestion in major Indian cities such as Mumbai, Bengaluru, and the Delhi NCR Region is a critical issue leading to significant delays, increased fuel consumption, higher emissions, and reduced road safety. The current fixed-timer traffic signal systems fail to adapt to real-time traffic conditions, causing inefficient traffic flow and exacerbating congestion during peak hours.

Context and Background:

With rapid urbanization and the increase in vehicle numbers, the traffic infrastructure in these cities is under severe strain. Traffic congestion not only impacts the daily lives of commuters but also affects emergency response times and overall urban mobility. Despite various measures, the traditional traffic management systems have not been able to address these challenges effectively.

Market Research

Root Causes for Traffic Congestion

Inadequacy of Traffic System

- **Overview:** India's traffic systems often lack advanced technologies. Outdated signals and lack of real-time monitoring are common.
- **Impact:** Inefficient signals and poor synchronization lead to frequent stops and delays, worsening congestion.

Narrow Roads

- **Overview:** Many roads in Indian cities are too narrow for current traffic volumes.
- **Impact:** Limited road width creates bottlenecks and choke points, slowing down traffic, especially during peak hours.

Illegal Parking

- **Overview:** Unauthorized roadside parking reduces available road space.
- **Impact:** Blocks lanes, forcing vehicles to slow down or stop, increasing congestion and delays.

Increasing Number of Vehicles

- **Overview:** The rapid rise in private vehicles has outpaced infrastructure growth.
- **Impact:** High vehicle density leads to overcrowded roads, frequent jams, and longer commute times.

No-Low Public Transport Facility

- **Overview:** Insufficient public transport options push more people to use private vehicles.
- **Impact:** Higher private vehicle usage adds to road congestion and pollution.

Construction and Maintenance

- **Overview:** Ongoing construction projects and poor road maintenance disrupt traffic flow.
- **Impact:** Temporary closures and diversions create delays and increase accident risks, worsening congestion.

Accidents and Road Conditions

- **Overview:** Frequent accidents and poor road conditions like potholes slow down traffic.
- **Impact:** Accidents and poor roads reduce capacity and create delays, contributing to congestion.

Improper Lane Management

- **Overview:** Lack of lane discipline and enforcement leads to chaotic vehicle movement.
- **Impact:** Disruptions in traffic flow and increased accident likelihood exacerbate congestion.

Repercussions of Traffic Congestion

Economic Costs

- In 2023, Bengaluru commuters lost an average of 132 hours due to traffic congestion, costing the local economy significantly. Similarly, Mumbai and Delhi commuters lost 92 and 81 hours, respectively
- Bengaluru alone suffers an annual economic loss of approximately ₹20,000 crore due to traffic congestion. The IT sector in Bengaluru incurs a loss of about ₹7,000 crore annually because of productivity loss from employees stuck in traffic
- Traffic congestion costs India around ₹1.44 lakh crore annually across major cities like Delhi, Mumbai, Bengaluru, and Kolkata

Environmental Impact:

- In cities like Delhi and Bengaluru, increased vehicle idling due to traffic congestion raises emission levels of harmful pollutants such as CO₂, NO_x, and particulate matter.
- For example, transportation accounted for 14% of global CO₂ emissions, and in India, the volume of CO₂ emissions from traffic has doubled between 1994 and 2007

Health Issues:

- Prolonged exposure to traffic congestion is linked to stress, higher blood pressure, and other health problems.

- The mental toll on commuters can be substantial

Loss of Lives

- Higher traffic density and frequent stop-and-go conditions increase the likelihood of accidents. Traffic congestion is a major factor contributing to the high number of road accidents in cities like Mumbai and Bengaluru.
- In 2021, India recorded over 120,000 deaths due to road accidents, many of which were exacerbated by traffic congestion

Repercussions of breaking Traffic Signal

Increased Congestion

- Without signals, intersections become a free-for-all, leading to indecisiveness and hesitation among drivers. This delays movement and creates bottlenecks, causing traffic to increase significantly.

Disregard for Lane Discipline

- In the absence of clear direction from signals, lane discipline often breaks down. Drivers try to squeeze in, leading to last-minute maneuvers and blocking the flow of traffic

Rise in Road Rage

- The frustration of slow movement and perceived unfairness can lead to aggressive driving behavior and road rage incidents which can create accident situations or road jam situations leading to increased congestion

Higher Risk of Accidents

- Without the designated right of way, the chances of collisions at intersections significantly increase which can in turn lead to personal life and also create multiple accident scenarios leading to reduced traffic flow in the area

Pedestrian Safety Issues

- Pedestrians crossing the road rely on signals for safe passage. Malfunctioning signals make them vulnerable to accidents

Incident Management Overview

Emergency Operations / Incident Management involves coordination among various city agencies through the emergency center. The Traffic Management Integrated Control Centers (TMICCs) play a supporting role in the following areas:

Detecting Incidents

- Receive alerts from various sources, including public reports.
- Verify incidents via surveillance feeds and share with emergency centers.
- Exchange incident information with emergency centers and allow control of surveillance equipment.

Coordination for Emergency Vehicles Dispatch

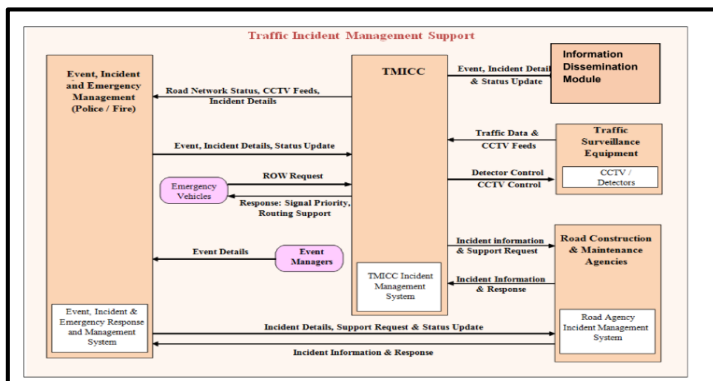
- Plan and coordinate incident response and recovery with emergency centers.
- Provide special routing and signal pre-emption for emergency vehicles.
- Exchange detailed incident information with relevant agencies and the media.
- Share road network status, including closures and alternate routes, with emergency and maintenance agencies.

Support During Evacuation

- Coordinate evacuation planning with emergency centers, including routes and timing.
- Support traffic control strategy adjustments to facilitate evacuation.
- Execute evacuation strategies by setting closures, detours, and establishing routes.

TMICCs Actions Based on Incident Nature

- Analyze the situation and take necessary actions such as adjusting traffic control strategies, coordinating with various agencies, and ensuring effective communication and information dissemination.



Government Statistics

Ministry of Road Transport and Highways

- **Total Allocation:** Rs 2,70,435 crore, a 25% increase from the previous year
- **Investment in NHAI:** 60% of the additional allocation, due to NHAI's decision not to borrow from the market
- **Private Investment:** Constituted only 7% of total investment in roads in 2020-21
- **Road Safety:** Rs 330 crore allocated for 2023-24, a 20% increase from the previous year
- **Underutilization:** 41% underutilization of road safety funds between 2017-18 and 2022-23

AMRUT and Smart Cities Mission

- **Revised Estimates for 2023-24:** Rs 13,200 crore
- **Interim Budget:** Rs 10,400 crore, a decrease of approximately 21%
- **Initial Budget Estimates for 2023-24:** Rs 16,000 crore
- Major cities like Gurgaon and Bangalore planned to install smart traffic lights.
- Gurgaon planned to have 91 smart traffic signals by January 2023, increasing to 111. Bangalore aimed to have 500 AI-based traffic signals in the 3 years

What are the other countries doing?

China Model

The ancient city of Xi'an, China, has successfully tackled its congestion issues by implementing an intelligent traffic management system using **Hikvision technology**. This system leverages AI-powered video technology to monitor and manage traffic flow effectively. Here are the key aspects of the solution:

Category	Details
Challenges	Ancient City Layout: Xi'an's 'checkerboard' layout and ancient walls restrict vehicle movement, causing congestion Growing Population: Population of over 10 million and numerous construction projects exacerbate traffic issues Preservation of Historical Sites: Numerous ancient ruins limit urban development and traffic expansion

Solutions	AI-Powered Cameras: Hikvision's Checkpoint Capture Cameras and Intersection Violation Capture Units monitor and record traffic violations in real-time Sonar Monitoring: Detects illegal use of car horns in restricted areas Visual Command and Dispatch Platform: Real-time video streams and intelligent mobile applications create a dynamic command and control center Predictive Analysis: Machine learning predicts congestion patterns for preemptive measures. Intelligent Signal Control: Dynamically adjusts traffic signal timing using real-time data
Benefits	Increased Traffic Throughput: Improved by 10% due to intelligent signal control Reduced Journey Time: Average vehicle journey time decreased by 12% Improved Driver Behavior: Traffic violations dropped by 30%, indicating better compliance with traffic rules Enhanced Incident Detection: Proactive warning functions increased incident detection rates by over 30%
Conclusion	Xi'an's intelligent traffic management system effectively addresses congestion, supports urbanization efforts, and preserves the city's historical heritage. This technology-driven approach creates a smarter and more efficient urban environment.

Japan Model

Tokyo, Japan, a bustling metropolis, successfully tackled its traffic congestion issues by implementing an intelligent traffic management system utilizing advanced technology. This system uses AI-powered video technology, data analytics, and real-time monitoring to manage traffic flow effectively.

Category	Details
Challenges	High Population Density: Tokyo's dense population creates heavy traffic, especially during peak hours. Complex Road Network: A complex and extensive road network, including highways, tunnels, and bridges, makes traffic management challenging. Frequent Natural Disasters: Earthquakes and typhoons frequently disrupt traffic and require rapid response and recovery.
Solutions	AI-Powered Cameras: High-resolution cameras installed at key intersections and along major roads monitor and record traffic violations and congestion in real-time. Real-Time Traffic Data Integration: Integration with Google Maps and other navigation services for real-time traffic updates and incident reports.

	Predictive Analysis: Machine learning models analyze historical and real-time data to predict congestion patterns and identify hotspots. Intelligent Signal Control: Traffic signals dynamically adjust timing based on real-time traffic conditions to optimize flow Emergency Response Integration: Real-time monitoring and automated incident reporting to quickly deploy emergency services during natural disasters.
Benefits	Reduced Commute Times: Average vehicle journey times decreased by 20%. Improved Traffic Flow: Traffic throughput improved by 15% due to intelligent signal control and real-time traffic management. Enhanced Public Safety: Rapid response to incidents and natural disasters, reducing the impact on traffic and ensuring public safety. Reduced Traffic Violations: Traffic violations dropped by 25% due to enhanced monitoring and enforcement.
Conclusion	Tokyo's intelligent traffic management system effectively addresses the city's traffic congestion, supports rapid urban growth, and enhances public safety during natural disasters. This technology-driven approach creates a smarter, more resilient, and efficient urban environment.

User Interviews

The answers to user interview are given by an ex-traffic DSP who is currently working as a DSP in the crime department

Usage of CCTV cameras

- The CCTV cameras are installed in all major signals and junctions of the metro cities
- The feed received used only for post incident surveillance. It is not used for real time monitoring
- Neither manual Live monitoring nor Automated live monitoring of the signals is done
- Enough manpower is not available to monitor the feed
- CCTV cameras don't work sometimes due to lack of maintenance and the government often delays in releasing funds
- Few major junctions in these cities have implemented ANPR(Automated Number Plate Recognition) where the offenders are captured by CCTV automatically, but the same is not available in majority of the junctions.

Incident Detection and Response

- There is no automated incident identification mechanism. The incident is detected through walkie talkie or if any commuter passes the info to police
- For planned events like processions, rallies, festivals etc, a thorough planning is done by the respective departments and necessary resources are deployed. If any traffic diversion is

required, the government officials releases a press statement regarding the same. Some govt bodies also update the same through twitter

Traffic Signal Regulation

- The timer based traffic signals are used but during peak hours, the police control it manually since there is no smart traffic signal which automatically manages the signal based on traffic density

Measures taken to combat traffic

- Immediate and easy to implement solution is to reduce the number of U turns which will reduce the traffic significantly
- Strict fines for wrongly parked vehicles and street hawkers to ensure they are not blocking the roads leading to traffic congestion

Problems

Inefficient Incident Management

- **Problem:** Incident detection and reporting currently rely on manual methods such as public reports and walkie-talkie communication. There is no automated system for promptly identifying and reporting incidents.
- **Impact:** Delays in incident detection and reporting lead to slower response times, prolonged traffic disruptions, and reduced efficiency in traffic management.

Underutilization of Traffic Management Technology

- **Problem:** Advanced traffic management technologies, such as CCTV and ANPR systems, are not fully utilized for real-time monitoring and incident management. Manual processes are heavily relied upon instead.
- **Impact:** The potential benefits of existing technologies are not fully realized, resulting in less efficient traffic management and delayed incident responses.

Inadequate Public Notification Systems

- **Problem:** Current methods of notifying the public about traffic incidents and diversions (e.g., press, social media) are insufficient and may not reach all commuters in a timely manner.
- **Impact:** Ineffective public notifications prevent timely dissemination of critical information, causing unpreparedness among commuters and exacerbating congestion and delays.

Inefficient Traffic Signal Management

- **Problem:** Most traffic signals operate on fixed-timer systems, requiring manual adjustments and lacking real-time adaptive control.
- **Impact:** Fixed-timer systems are inefficient during peak hours and incidents, leading to suboptimal traffic flow and increased congestion.

User Persona

Name	Rajesh Sharma	Meena Patel	Anjali Deshmukh	Suresh Reddy
Role	Traffic Control Centre Operator	Field Traffic Police Officer	Public Transport Planner	Emergency Services Dispatcher
Age	35	28	45	50
Responsibilities	➤ Monitor traffic flow and incidents ➤ Coordinate with traffic police and emergency services ➤ Manage real-time traffic data and incident reports	➤ Direct traffic at busy intersections ➤ Respond to traffic incidents and accidents ➤ Ensure compliance with traffic regulations	➤ Plan and optimize public transport routes ➤ Coordinate with traffic control centres and city planners ➤ Enhance efficiency and coverage of public transport systems	➤ Coordinate emergency response efforts ➤ Communicate with traffic control centres and emergency teams ➤ Manage incident data and response logistics

Challenges	➤ Inconsistent and delayed incident reporting ➤ Underutilized traffic management technologies ➤ Inefficient communication and resource allocation	➤ Lack of real-time incident alerts ➤ Inadequate traffic signal control systems ➤ Insufficient public notification about traffic diversions	➤ Insufficient data on traffic patterns and congestion ➤ Coordination issues with traffic management and city planning ➤ Limited resources for expanding public transport infrastructure	➤ Delays in incident detection and response coordination ➤ Inadequate real-time traffic and incident data ➤ Inefficient routing for emergency vehicles
Goals	➤ Improve incident detection and response times ➤ Enhance utilization of traffic management technologies ➤ Streamline communication and resource management	➤ Receive timely alerts for incidents and accidents ➤ Utilize adaptive traffic signal systems ➤ Improve public communication regarding traffic changes	➤ Obtain detailed traffic data for better planning ➤ Improve coordination with traffic management systems ➤ Enhance public transport infrastructure and services	➤ Improve incident detection and reporting systems ➤ Access real-time traffic and incident data for better coordination ➤ Optimize routing for emergency vehicles to reduce response times

Solutions

Solution Prioritization

High Priority	Medium Priority	Low Priority
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➤ Automated Incident Management ➤ Real time alerts to notify incidents' occurrence and clearance ➤ Tow truck Route optimization ➤ Temporary one way routing suggestions ➤ Smart Automated Signal Handling based on Traffic density ➤ Multilingual option for the user to select as a preference	➤ Road widening suggestions ➤ U-Turn closure/Opening suggestions ➤ Stand By Ambulance and Fire Truck suggestions ➤ Smart Parking management	➤ Traffic Personnel management
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Non-Recurring Traffic:

Automated Incident Reporting

- **Google Maps Data Integration**

Description: Utilize real-time traffic data from Google Maps, including traffic density, average speeds, and incident reports.

Implementation Steps:

Data Collection:

- **Google Maps API:** Use the Google Maps Traffic Layer API to collect real-time traffic data. This includes:
 - Traffic density (measured through congestion levels on different road segments).
 - Average speeds of vehicles on various road segments.
 - Incident reports such as accidents, road closures, and construction.

Data Analysis:

- **Pattern Recognition:** Implement machine learning models to analyze traffic patterns. This can include:
 - **Historical Data Analysis:** Compare current data with historical data to identify unusual patterns.
 - **Congestion Prediction:** Use time-series forecasting models (e.g., ARIMA, LSTM) to predict future congestion.
- **Hotspot Identification:** Identify congestion hotspots by analyzing areas with consistently high traffic density and slow speeds

Integration:

- **Data Fusion:** Integrate Google Maps data with other data sources (e.g., social media, road cameras) for a comprehensive traffic management system. Use a central data warehouse or a real-time data processing system like Apache Kafka or Apache Flink.

● Social Media Scraping

Description: Use bots to scrape social media platforms like X (formerly Twitter), Instagram, and Snapchat for hashtags, images, and posts related to traffic conditions.

Implementation Steps:

Data Collection:

- **Social Media APIs:** Use APIs from social media platforms to collect posts containing relevant hashtags and keywords related to traffic. For platforms without APIs, develop web scraping bots.

Data Processing:

- **Natural Language Processing (NLP):** Use NLP techniques to analyze text data. This includes:
 - **Entity Recognition:** Identify mentions of locations, incidents, and traffic conditions.
 - **Sentiment Analysis:** Gauge public sentiment to identify areas of high concern.
- **Computer Vision:** Analyze images and videos using computer vision techniques to identify traffic conditions and incidents

Sentiment Analysis:

- **Public Perception:** Perform sentiment analysis to gauge public perception of traffic conditions. Identify areas where the sentiment is particularly negative, indicating significant traffic issues. The identification can be done by analyzing the posts on social media

- **Cameras on Roads and Body**

- **Description:** Utilize road and body cameras for real-time traffic monitoring and enforcement
- **Implementation:**
 - **Real-Time Monitoring:** Use AI-powered video analytics to monitor traffic violations, incidents, and congestion in real-time
 - **Data Storage and Analysis:** Store video data for historical analysis and integrate it with other data sources

Manual Incident Reporting

Description: Police personnel can manually update incidents in the app, including screenshots and required resources. This information is pushed to the traffic control room in real-time.

Implementation Steps:

- **Incident Reporting:**
 - **User Interface:** The mobile application should have an intuitive and user-friendly interface, allowing police personnel to report incidents quickly.
 - **Forms and Fields:** Include fields for entering incident details such as location, time, type of incident (e.g., accident, roadblock), severity, and any other relevant information.
 - **Image Upload:** Provide an option to capture and upload images directly within the app. This includes taking photos of the incident site, vehicles involved, and any other pertinent details.
- **Real-Time Data Upload**
 - **Connectivity:** Ensure the mobile application is always connected to the central server, either through cellular data or Wi-Fi, to facilitate real-time data upload.
 - **Data Format:** Structure the data in a JSON format for easy transmission and processing.
- **Image Processing and Analysis**
 - **Real-Time Upload:** Images captured through the mobile app should be uploaded to the server instantly to avoid manual overhead.
 - **Computer Vision:** Use computer vision techniques to analyze uploaded images for incident verification, documentation, and further analysis.
- **Integration with Central Server**
 - **Real-Time Data Processing:** The central server should process the incoming data and store it in a structured database. Use cloud services for scalability and reliability.
 - **Database Design:** Design a robust database schema to handle incident reports, images, and related metadata.
- **Integration with Traffic Control Room**

- **Real-Time Updates:** Push real-time updates to the traffic control room dashboard for immediate action. Use WebSockets for real-time communication.
- **Data Visualization:** Develop a dashboard to visualize incident reports, including maps showing incident locations, incident details, and status updates.

Suggestions Based on the Algorithm:

- **Criticality:**

- **Critical:**

- **Description:** Incidents that require immediate attention and rapid response.
- **Examples:** Major accidents, road closures due to severe weather or construction, significant traffic jams, high congestion hotspots, etc.
- **Actions:** Immediate dispatch of police and emergency services, real-time notifications to the control room, and priority handling on the dashboard.

- **High:**

- **Description:** Incidents that are serious but not life-threatening, and need timely intervention.
- **Examples:** Minor accidents, moderate congestion, temporary road obstructions, etc.
- **Actions:** Dispatch of traffic police or tow trucks, notifications to the control room, and high priority on the dashboard.

- **Medium:**

- **Description:** Incidents that are less severe but still require monitoring and possible action.
- **Examples:** Slow traffic due to minor construction, small-scale road repairs, minor violations, etc.
- **Actions:** Monitor the situation, potential dispatch of resources if conditions worsen, and medium priority on the dashboard.

- **Low:**

- **Description:** Minor incidents that do not require immediate action but should be logged for record-keeping and analysis.
- **Examples:** Slight traffic slowdowns, routine traffic violations, public complaints, etc.
- **Actions:** Log the incident for future reference, low priority on the dashboard.

- **Intimate Necessary Government Bodies**

- **Real-Time Alerts:** Set up a system to send real-time alerts to traffic management authorities, police departments, and emergency services
- **Automated Reports:** Generate automated reports highlighting key insights, trends, and recommendations, and send them to relevant government bodies
- **Incident Response:** Implement a protocol for immediate response to traffic incidents, including the automated dispatch of traffic police and emergency services

- **Advise on the Number of Resources Required**

- **Field Personnel:** Information to be sent to the control room and suggest the number of personnel required to handle the incident

Incident Notification System

Assumption: Availability of location data for all vehicles and traffic management systems

- **Notification Triggers:**

- These triggers will be activated only in case critical incidents as mentioned above
- Set up triggers based on incident severity and location. Notification to be sent out only for the Major incidents
- Notification to alert users within the defined range of 3kms

- **Clearance Notifications:**

- Continuously monitor the incident site using the same data sources
- Trigger notifications when data indicates the incident has been cleared

Recurring Traffic:

Smart Signal Handling and Traffic Management

- Major traffic areas to have cameras and integration with Google Maps is done via API

Traffic Data Collection and Integration:

- **Camera Monitoring:**

- Utilize available cameras on roads to monitor real-time traffic conditions, including congestion levels and vehicle movement patterns.

- **Google Maps Integration:**

- Integrate Google Maps APIs to gather traffic congestion data for specific routes, narrow lanes, and U-turn points.

- **Data Aggregation:**

- Collect and aggregate data from cameras and Google Maps to create a comprehensive traffic profile for analysis.

Traffic Analysis and Pattern Recognition:

- **Machine Learning Algorithms:**

- Machine learning models to analyze historical and real-time traffic data.

- **Pattern Recognition:**

- Identify recurring congestion points, peak traffic hours, and typical traffic flow patterns.

- **Predictive Analysis:**

- Use predictive analytics to forecast future traffic conditions based on historical trends and current data.

Smart Signal Handling Strategy:

Algorithms will take the data into account and then provide the dynamic signal timing.

At a junction, instead of using the same timer for all four side, peak traffic hours will have more time to the higher traffic flow side. This will change as the traffic will normalise again.

- **Dynamic Signal Timing:** Implement dynamic signal timing adjustments based on real-time traffic data.
 - **Traffic Density:** Adjust signal timings based on current traffic density to minimize congestion and maximize throughput.
 - **Congestion Hotspots:** Prioritize green lights for directions with higher congestion to alleviate traffic buildup.
 - **Time-of-Day Adjustments:** Modify signal timings according to peak traffic hours identified through data analysis.
- **Adaptive Algorithms:**
 - Utilize adaptive algorithms that continuously learn and adjust based on real-time feedback from cameras and Google Maps data.

Temporary One-Way Traffic Management:

- **Identification of Hotspots:**
 - Traffic data analysis to identify locations prone to severe congestion during specific times.
- **Peak Period Management:**
 - Temporary implementation of one-way traffic flows during peak periods to manage traffic effectively.
 - **Morning and Evening Peaks:** Designate one-way routes in the morning towards major business areas and reverse in the evening to ease rush-hour traffic.
 - **Event-based Adjustments:** Modify traffic flows dynamically during events, roadwork, or emergencies to optimize traffic movement.

Implementation and Monitoring:

- **Traffic Control Center:**
 - Establish a centralized traffic control center equipped with real-time monitoring screens and analytics dashboards.
- **Automated Alerts:**

- Set up automated alerts for traffic anomalies, system malfunctions, or emergency situations requiring manual intervention.
- **Continuous Optimization:**
 - Regularly review and optimize smart signal handling strategies based on performance metrics and user feedback.

Resource Management:

- **Dynamic Deployment:**
 - Algorithms that dynamically allocate traffic police based on real-time and predictive traffic data that is available with the control room
- **Event-Based Deployment:**
 - Allocate additional personnel during special events, road closures, or emergencies
- **Shift Planning:**
 - Optimize shift schedules to ensure adequate coverage during peak traffic hours and critical periods

Tow Truck Route Optimization:

One of the challenges faced is illegal parking and this causes a huge congestion if wrongly parked during the peak hours

- **Dynamic Routing:**
 - Algorithms will dynamically adjust tow truck routes based on real-time traffic conditions and parking violations
- **Priority-Based Dispatch:**
 - Prioritize tow truck deployment to areas with severe congestion or high traffic violations. Tow trucks will be moving throughout the city so the placement will be such that they are around the areas of peak traffic
- **Optimized Sequencing:**
 - Sequence tow truck visits to maximize efficiency and minimize travel time between pickup locations

Smart City Planning:

It is impossible to increase the size of the urban area. Technology is the only way through which we can allow the modern city to grow and develop.

We won't be able to change the infrastructure but we can surely recommend the changes that if taken can benefit a lot in traffic management.

Dashboard will be available to the government bodies to see the real time impact of the change.

Components of the Dashboard:

Real-Time Traffic Data Monitoring:

- **Traffic Density:**
 - Live traffic density maps indicating congestion levels.
- **Incident Reports:**
 - Real-time data on traffic incidents, accidents, and road closures.
- **Average Speeds:**
 - Heatmaps show average speeds across different road segments.

Historical Data Analysis of the traffic collected through google maps integration and camera feed:

- **Traffic Trends:**
 - Analysis of historical traffic patterns to identify recurring congestion hotspots.
- **Accident Data:**
 - Historical accident data to identify high-risk areas.
- **Road Usage Statistics:**
 - Volume and types of vehicles using specific road segments over time.

Commented [1]: headings are vague, Historical data analysis for ??
by reading the headings of high level solution I should get the idea of solution

Predictive Analytics to forecast the traffic:

- **Congestion Forecasting:**
 - Predict future congestion patterns based on historical and real-time data.
- **Growth Projections:**
 - Traffic growth projections based on urban development plans and population growth trends.

Current Infrastructure Assessment:

- **Current Road Capacity:**
 - Analysis of current road capacities and their utilization levels.
- **Condition Reports:**
 - Assessment of road conditions and maintenance needs.
- **Bottleneck Identification**
 - Identification of infrastructure bottlenecks causing major traffic delays.

Recommendation Engine:

- **New Lanes:**
 - Suggestions for adding new lanes to existing roads where congestion is highest.
 - These suggestions are given keeping the land acquisition in mind

- **Wider Roads:**
 - Recommendations for widening roads based on traffic volume and road usage data.
 - There could be illegal construction which could cause narrow roads so system will be able detect that and throw the suggestion
- **Alternative Roads:**
 - Recommendations for U-turn to be provided, this will include both the number of U-turns and the placement of U-turns
- **Overpasses and Flyovers:**
 - Identification of critical intersections where overpasses or flyovers could significantly reduce congestion.
- **Alternative Routes:**
 - Proposals for new road segments to divert traffic from congested areas.



Visualization and User Interface:

- **Interactive Maps:**
 - Interactive maps showing current traffic conditions, proposed infrastructure projects, and their impact.
- **Data Filters:**
 - Filters to view data by time, location, type of road, and traffic conditions.
- **Scenario Analysis:**
 - Tools to simulate the impact of proposed infrastructure changes on traffic flow.

User Flows

Incident management - Traffic Control Tower

<https://whimsical.com/control-room-user-flow-for-incident-management-YPFHFTYhF2zTXww7wTv9cM>

Incident Management - On Ground Traffic Police

<https://whimsical.com/b2b-app-user-flow-for-traffic-congestion-management-GSDPX8BPL9mAEdH6EsCji9>

Wireframes

Traffic Havalдар

<https://www.figma.com/proto/C8DsOHC6X0fHaguBo7weRV/Untitled?node-id=0-1&t=eEiOOXvNiWl8dsnR-0&scaling=scale-down&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=64%3A161>

Traffic Control Room Operator flow

<https://www.figma.com/proto/VLDHjvxPwGwtelsMgQ7NQb/Untitled?page-id=0%3A1&node-id=1-204&viewport=793%2C541%2C0.19&t=jHmhst1QxEZcmz4B-1&scaling=min-zoom&content-scaling=fixed>

Recurring Traffic

<https://www.figma.com/proto/C8DsOHC6X0fHaguBo7weRV/Untitled?page-id=0%3A1&node-id=75-378&viewport=199%2C-2115%2C0.45&t=sm59Z1JKFyXG1ZyP-9&scaling=scale-down&content-scaling=fixed&starting-point-node-id=75%3A378&show-prot-sidebar=1>

Non Recurring Traffic

<https://www.figma.com/proto/C8DsOHC6X0fHaguBo7weRV/Untitled?page-id=0%3A1&node-id=25-275&viewport=199%2C-2115%2C0.45&t=bk9XSziCi8GIAGVk-9&scaling=scale-down&content-scaling=fixed&starting-point-node-id=25%3A275&show-prot-sidebar=1>

Metrics

Core Parameters	Metrics
Incident Management	• % of incidents reported manually (Traffic policemen) • % of incidents reported successfully via API data (Google, SM, cameras) • % of false positive incident reporting (Automated reporting)

	● SLA % for Incident Resolution ● % of successful incident image uploads ● % of Critical incidents ● TAT for incident resolution ● % of Weekly, Monthly reported incidents
Signal handling	● Average waiting time at a signal ● % of signals not working/ static signals ● Deviation of Predicted vs. Actual Signal Time at Peak Hours
Routing	● No. of areas identified as congestion hotspots ● % of roads diverted to one-way roads ● No. of new traffic policemen diverted to congestion hotspots against existing traffic personnel in the area